

PAPER IMPROVEMENT

(Since the original code was not provided, the code I used for testing was the one I recreated myself, and I conducted certain code tests based on the article. Please refer to *Result Reproduction.pdf* for details.)

The original assumption:

For the j -th site, we assume its distribution:

$$f_j(y_{ij}; \theta_j) = \lambda_j f(y_{ij}; \mu_1) + (1 - \lambda_j) f(y_{ij}; \mu_0)$$

This assumption is somewhat too strong because it is difficult for us to ensure that each site has the same distribution. This distribution may be affected by different donors, different sample information, and even batch effects, resulting in a certain degree of deviation. For more details, please refer to the document (Offset Experiment.pdf).

So here we make a certain degree of relaxation, allowing each site to have its own unique parameters. We assume:

$$f_j(y_{ij}; \theta_j) = \lambda_j f(y_{ij}; \mu_1 + b_{1j}) + (1 - \lambda_j) f(y_{ij}; \mu_0 + b_{0j})$$

The parameter configuration for each site is as follows: shared parameters + site offset (which may be determined by different donors, different sample information, or even batch effects)

In this situation, we can improve the distributed EM of the article.

For the Q function, the original text intends to anchor a lead site and use the information from other sites for supplementary calculation, and only perform gradient information transmission rather than the original data. That is to say, it constructs:

$$\tilde{Q}(\mu | \theta^t) = \check{Q}(\mu^t | \theta^t) + \left\langle \nabla_{\mu} Q_{\mu}(\mu^t | \theta^t) - \nabla_{\mu} \check{Q}(\mu^t | \theta^t), \mu \right\rangle, \quad (1)$$

Here, $\check{Q}$ is calculated solely using the density ratio and the internal data of the lead site, without requiring any other information.

We construct each term by following the pattern (1).

Selected lead site:

$$\begin{aligned} \check{Q}(\mu | \theta^t) &= \frac{1}{Kn} \sum_{j=1}^K \sum_{i=1}^n t(y_{i1}, \eta_j^t) \left[w_{\theta^t}^j(y_{i1}) \log(f(y_{i1}; \mu_1 + b_{1j})) + (1 - w_{\theta^t}^j(y_{i1})) \log(f(y_{i1}; \mu_0 + b_{0j})) \right] \\ t(y_{i1}, \eta_j^t) &= \frac{\lambda_j^t f(y_{i1}; \mu_1 + b_{1j}) + (1 - \lambda_j^t) f(y_{i1}; \mu_0 + b_{0j})}{\lambda_1^t f(y_{i1}; \mu_1 + b_{11}) + (1 - \lambda_1^t) f(y_{i1}; \mu_0 + b_{01})} \\ \eta_j^t &= (\mu_0^t, \mu_1^t, \lambda_1^t, \lambda_j^t)^\top \end{aligned}$$

In the scenario we have set, we specify the parameters as follows:

$$\begin{aligned}
\mu &= (\mu_0, \mu_1) \\
b &= (b_{01}, b_{11}, b_{02}, b_{12} \dots b_{0K}, b_{1K}) \\
b_j &= (b_{0j}, b_{1j}) \\
\theta_j &= (\lambda_j, \mu, b_j) \\
\Lambda &= (\lambda_1, \dots, \lambda_K)
\end{aligned}$$

So, by updating the formula, we can divide it into two steps, EM, and then organize it accordingly:

E step: update confidence

$$w_{\theta_j}^j(y) = E(Z \mid \theta_j, y) = \frac{\lambda_j f(y; \mu_1 + b_{1j})}{\lambda_j f(y; \mu_1 + b_{1j}) + (1 - \lambda_j) f(y; \mu_0 + b_{0j})}.$$

M-step: update parameters μ, b, λ

First, we need to update parameters within each sub-site, that is:

$$\tilde{\lambda}_j^{t+1} = \frac{1}{n} \sum_{i=1}^n w_{\theta^t}^j(y_{ij})$$

For local shift, what I am doing here is to solve the problem using optimization, that is, to calculate within the sub-site:

$$\min_{\delta_{1j}} \sum_i w_{\theta^t}^j(y_{ij}) \|y_i - \mu_1 - b_{1j}\|_{\Sigma^{-1}}^2 + \rho \|b_{1j}\|^2$$

Here, ρ is a parameter that ensures b does not deviate completely. Appropriate restrictions are imposed, and a closed-form solution can be obtained:

$$\begin{aligned}
b_{1j} &= (N_1 \Sigma^{-1} + \rho I)^{-1} \Sigma^{-1} (S_1 - N_1 \mu_1). \\
N_1 &= \sum_i w_{\theta^t}^j(y_{ij}) \\
S_1 &= \sum_i w_{\theta^t}^j(y_{ij}) y_{ij}
\end{aligned}$$

Similarly, b_{0j} can also be solved like that

For global mean μ , we updated each sub-site based on the original logic after applying the offset correction:

$$\tilde{\mu}^{t+1} = \arg \max_{\mu} \tilde{Q}(\mu \mid \theta^t).$$

Algorithm: Shift-Distributed EM

1. **Input:** site data $\{Y_j\}_{j=1}^K$, initial estimates $\mu^0, \lambda_j^0, b_j^0$;
2. **Initialize with** $\tilde{\mu}^0 = \mu^0$;
3. **From** $t = 0$ **iterate until convergence:**
4. **In Site** $j = 1$ **to** $j = K$
5. Compute local responsibilities and update $\tilde{\lambda}_j^t, b_{0j}^t, b_{1j}^t$;
6. Transfer $\tilde{\lambda}_j^t, b_{0j}^t, b_{1j}^t$, and $\nabla_{\mu} Q_j(\tilde{\theta}_j^t | \tilde{\theta}_j^t)$ to Site 1;
7. **In Site 1**
8. Construct $\tilde{Q}(\mu | \tilde{\theta}^t)$ using density-ratio weighting and $\{\nabla_{\mu} Q_j(\tilde{\theta}_j^t | \tilde{\theta}_j^t)\}_{j=1}^K$;
9. Obtain $\tilde{\mu}^{t+1}$ by solving $\nabla_{\mu} \tilde{Q}(\mu | \tilde{\theta}^t) = 0$ and broadcast $\tilde{\mu}^{t+1}$;
10. **Output:** $\tilde{\theta}$.

For the simulation experiment, please refer to the file Offset Experiment.pdf.